Data Analytics and Visualization

Module - 01

Introduction to Data Analytics: Data Analytics: Definition, importance, and applications in industry; Types of analytics: descriptive, diagnostic, predictive, prescriptive; Data analytics lifecycle: problem definition, data collection, analysis, deployment; Types of data: structured, semi-structured, unstructured; Time Series Data; Data sources and formats; Python for data analytics: NumPy, Pandas, reading CSV/JSON files, indexing, filtering, grouping and aggregation; Overview of Big Data: characteristics (5Vs), ecosystem and tools.

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Introduction to Data Analytics:

What is Data Analytics?

“Suppose you are the owner of a supermarket. You have sales data for the last 1 year. What can you do with this data?”

From the sales data, we can find:

- Find which product sells the most
- Find which month has more sales
- Find which products are not selling
- Find which customers buy more
- Decide how much stock to purchase

“All these activities involve looking at data, finding useful information from it, and using that information to make better decisions. This process is called Data Analytics.”

Definition of Data Analytics:

Data Analytics is the process of collecting, cleaning, analyzing, and interpreting data to find useful information and support better decision-making.

(OR)

Data Analytics = Looking at data to understand what is happening and make better decisions.

Example:
Suppose a college has student attendance data:

Student	Attendance
A	92%
B	85%
C	62%
D	95%
E	58%

We can analyze this data to answer:

- Who has low attendance?
- What is the average attendance?
- How many students have less than 75% attendance?
- Which class has the highest attendance?
- Is attendance improving or decreasing?

The raw attendance numbers themselves are **data**.

After analyzing them, we obtain **useful information**.

For example:

"30% of students have attendance below 75%." **This statement is no longer just raw data.** That is an **insight** obtained from data.

Data vs Information vs Insight

This is an important concept :

1. Data

Data is raw facts and values that are collected.

Example:

85, 72, 91, 65, 55, 88, 92

By themselves, these numbers don't tell us much.

Note:
Data = What we have
Information = What the data tells us
Insight = What we understand from it

2. Information

When we organize and process the data, we get information.

For example:

Average attendance = 78%

Now the numbers have meaning.

3. Insight

An insight is a useful conclusion obtained from analyzing information.

For example:

"Students whose attendance is below 70% are more likely to have difficulty completing laboratory work."

This insight can help the college take action.

Simple flow is :
Data → Analysis → Information → Insight → Decision

Data

Raw values.

↓

Analysis

We examine the data.

↓

Information

We obtain meaningful results.

↓

Insight

We understand what those results mean.

↓

Decision

We decide what action should be taken.

Example

Attendance data

↓

Analyze attendance

↓

30% students below 75%

↓

Students with low attendance are at risk

↓

College decides to provide counseling

"The ultimate purpose of Data Analytics is not just to calculate numbers. It is to use those numbers to make better decisions."

Why Do We Need Data Analytics?

"Suppose our college has the attendance and marks of 5,000 students. Is simply storing all this data useful?" – **Yes**

"Can the college easily look at 5,000 students one by one and identify which students are having attendance problems?" – **No**

"This is where Data Analytics becomes important. We use analytics to examine large amounts of data, find useful patterns, identify problems, and make better decisions."

Main Idea:

Storage gives us data. Analytics helps us understand that data.

Organizations Generate Data Every Day

Almost everything we do today creates data.

For example, when you:
- Use Google
- Make a UPI payment
- Order food
- Watch YouTube
- Buy something online
- Use Instagram
- Book a ticket
- Attend an online class
- Use a banking application some form of data is generated.

So, we can say:
Modern organizations are data-rich.
But being **data-rich does not automatically mean being information-rich**.
They need analytics to convert the data into useful information.

For example, when you make a UPI payment:
The system may record:
- Amount
- Date
- Time
- Sender
- Receiver
- Transaction status

All this becomes **data**.

Modern organizations are data-rich, but being data-ri
does not automatically mean being information-rich.

Having 1 crore records doesn't mean we understand thos
records.
We need **analytics** to convert data into useful information

1. College Example

A college may have **5,000 students**.
Every semester, the college generates a huge amount of data.

Student data

- Student ID
- Name
- Department
- Semester
- Attendance
- IA marks
- Assignment marks
- Lab marks
- End-semester marks

So, the college database can contain **thousands of records**.

Next question is **“What can we do with this data?”**

What happens if we only store the data?
Suppose the college has this:

Student	Attendance	IA Marks	Final Marks
Ravi	92%	82	88
Priya	85%	78	81
Arun	62%	45	51
Sneha	95%	90	94
Kiran	58%	40	46

This table is **raw data**.

The college may want to know:
- How many students have attendance below 75%?
- Which students are at academic risk?
- Which subject has the lowest average marks?
- Which department has the highest pass percentage?
- Is attendance related to academic performance?
- Which students need additional support?

Simply storing the data doesn't answer these questions.
Analysis is required.

Data Analytics Converts Data Into Information

Suppose we analyze the attendance data.

We find:

25% of students have attendance below 75%.

Now we have useful information.

We analyze marks and discover:

Students with low attendance generally have lower average marks.

Now we have an **insight**.

The college can then take action:

Provide academic counseling and additional support to students with low attendance.

So, the flow is:

Raw Data
↓
Data Analytics
↓
Information
↓
Insight
↓
Decision
↓
Action

2. Example: Bank

Every day, a bank may process **millions of transactions**.

For example:

- ₹ 500 UPI payment
- ₹ 2,000 ATM withdrawal
- ₹ 1,500 online purchase
- ₹ 20,000 bank transfer
- ₹ 50,000 cash withdrawal

All these transactions generate data. But the bank doesn't just want to store them.

The bank wants to know:

- Which customers are actively using the bank?
- Which customers are not using their accounts?
- Which transactions are unusual?
- Which customers may be likely to take a loan?
- Are there suspicious transactions?
- Which services are most popular?

Data Analytics helps answer these questions. Where, Analytics helps banks understand transaction data and customer behavior

3.Fraud Detection Example

Suppose a customer's normal transactions look like this:

₹ 500
₹ 1,000
₹ 2,500
₹ 800
₹ 1,500 Suddenly: ₹ 5,00,000 is transferred from the account**.** **“Is this transaction automatically fraud?”**

The bank's analytics system can identify that this transaction is very different from the customer's usual behavior. It can flag it for further investigation.

So:

Transaction Data
↓
Analytics
↓
Unusual Pattern Detected
↓
Possible Fraud
↓
Bank Investigates

This is one reason Data Analytics is extremely important in banking.

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4. E-Commerce Example

Now take an online shopping website.
Every time you use an e-commerce application, many activities generate data.

For example:
You search for:
“Laptop”

Then you:
1.View a laptop.
2.Compare two laptops.
3.Add one to cart.
4.Remove it.
5.Search for another laptop.
6.Finally purchase one.

Every action generates data.

For example:

- What did you search?
- Which product did you view?
- How long did you view it?
- Did you add it to cart?
- Did you purchase it?

“Do you think Amazon/Flipkart only stores your purchase?”
No.
There are many activities before the purchase.
Those activities generate useful data.

Why Is This E-Commerce Data Useful?

The company can analyze this data to understand:

- **Customer behavior**

What products are customers interested in?

- **Product popularity**

Which products are searched most frequently?

- **Purchasing behavior**

Which products are actually purchased?

- **Customer preferences**

What type of products does a particular customer prefer?

- **Problems**

Why are customers adding products to the cart but not purchasing them?

These insights can help the company improve its business.

“The company is not just collecting customer activity. It is analyzing that activity to understand customers and improve the business.”

Importance of Data Analytics

There are several reasons why Data Analytics is important.

1. Better Decision-Making

- Data Analytics helps organizations make decisions based on **evidence rather than guesswork**.

Example

Suppose a restaurant wants to decide which food items to promote.

- Without analytics:

  "I think pizza is the most popular."

- With analytics:

Item	Monthly Sales
Pizza	1,200
Burger	800
Pasta	500
Sandwich	300

Now the restaurant has evidence.
The restaurant can clearly see that pizza has the highest sales.

So, it may decide to:

- Promote pizza
- Keep more pizza ingredients
- Introduce new pizza varieties

Data Analytics helps us replace "I think" with "Data sho

Raw purchase data → Analyze the data → Find a pattern → Give recommendations

2. Identifying Trends and Patterns

Data Analytics helps us identify patterns that may not be obvious by simply looking at raw data.

Example: Online shopping

Suppose an e-commerce company analyzes purchases

The company has lots of purchase data:

- Customer 1 → Mobile phone + Phone case
- Customer 2 → Mobile phone + Phone case
- Customer 3 → Mobile phone + Phone case
- Customer 4 → Laptop + Laptop bag

It discovers:
Customers frequently buy mobile phones together with phone cases.
This is a **pattern**.

The company can then recommend:
"Customers who bought this phone also bought this phone case."

This is an example of analytics being used in recommendation systems.

3. Understanding Customers

Companies want to understand:

- What customers like
- What they dislike
- What they purchase
- How often they purchase
- Why they stop using a product

Example

A streaming platform analyzes what users watch.

Suppose a user frequently watches:

- Action movies
- Thriller movies
- Crime series

The system can recommend similar content.

So, **“Analytics helps companies understand what customers are interested in and provide personalized services.”**

4. Improving Business Performance

Analytics can identify areas where an organization is performing well and areas that need improvement.

Example
Suppose a company has five branches:

Branch	Sales
Bangalore	₹ 20 lakh
Mysore	₹ 15 lakh
Chennai	₹ 25 lakh
Hyderabad	₹ 18 lakh
Pune	₹ 10 lakh

The company can investigate:
Why is Pune's performance lower?

Maybe:
- Less advertising
- Fewer customers
- Higher prices
- Poor location
- Less product availability

“Analytics helps us identify not only WHAT is happening, but also helps us investigate WHY it is happening.”

5. Predicting Future Outcomes

Data Analytics can also help organizations predict what may happen in the future.

For example, a company may analyze previous sales:
2023 → ₹ 10 lakh
2024 → ₹ 13 lakh
2025 → ₹ 17 lakh

Based on historical patterns, the company can estimate future sales.
This is called **predictive analytics**.

"We use past data to understand the present and sometimes predict what may happen in the future."

6. Detecting Fraud

Data Analytics is heavily used by banks and financial institutions.

Example

Suppose a person's normal transactions are:
₹ 500
₹ 1,200
₹ 800
₹ 2,000
Suddenly, there is a transaction:
₹ 5,00,000 from a different location.

An analytics system can identify this as unusual behavior and flag the transaction.

This can help detect:

- Credit card fraud
- Banking fraud
- Suspicious transactions
- Account misuse

"The important point is that analytics can learn or identify normal transaction behavior and highlight unusual patterns."

7. Reducing Costs

Analytics can help organizations identify unnecessary expenses.

Example

A manufacturing company analyzes electricity usage.

It discovers that one machine consumes significantly more electricity than others.

The company can investigate:

- Is the machine old or inefficient?
- Does it require maintenance?
- Can operating time be reduced?

If the company identifies and fixes the problem, it can **reduce electricity consumption and operational costs**.

“Analytics helps organizations find where money or resources are being unnecessarily spent.”

Applications of Data Analytics in Industries

1. Healthcare

"In hospitals, a lot of data is generated about patients, treatments, medicines, test results, etc."

Data Analytics can help hospitals:

- Understand patient records
- Identify diseases and risks
- Analyze treatment results
- Manage hospital resources

Simple example:

"Suppose a hospital analyzes patient data and finds that many patients have a particular disease during a particular season. The hospital can prepare medicines, doctors and beds in advance."

So: **Healthcare → Patient data → Analysis → Better healthcare decisions**

2. Banking and Finance

“Banks handle thousands or millions of transactions every day. Data Analytics helps banks understand these transactions.”

It can be used for:

- Detecting unusual transactions
- Understanding customer spending
- Assessing loan applications
- Managing financial risks

Example:

“Suppose a person's account normally has small transactions, but suddenly there is a very large unusual transaction. Data Analytics can help the bank identify this unusual behavior.”

3. Retail and E-Commerce

“Online shopping companies collect data about what customers search for, what they buy and what they don't buy.”

Data Analytics helps them:

- Understand customer preferences
- Recommend products
- Identify popular products
- Manage inventory
- Plan discounts

Example:

“If many customers buy mobile phones along with phone cases, the company can recommend a phone case when someone buys a mobile.”

4. Education

“Schools and colleges also generate a lot of data.”

For example:
- Marks
- Attendance
- Assignment scores
- Exam results
- Student performance

Data Analytics can help teachers:
“Suppose a teacher analyzes students' marks and notices that many students are performing poorly in one particular topic. The teacher can identify that topic and explain it again.”

So , analytics helps in **understanding student performance and improving teaching.**

5. Entertainment

“Streaming platforms such as Netflix and YouTube also use Data Analytics.”

They analyze:

- What users watch
- How long they watch
- What they search for
- What they skip

Then they can:

“Recommend movies, videos or shows that the user may like.”

User watches videos → Platform collects data → Analytics finds interests/patterns → Platform recommends similar videos

Types of Data Analytics

1. **What happened?** → Descriptive Analytics
2. **Why did it happen?** → Diagnostic Analytics
3. **What is likely to happen?** → Predictive Analytics
4. **What should we do?** → Prescriptive Analytics

1. Descriptive Analytics ("Descriptive Analytics summarizes past and present data to tell us what happened.") (OR)
Descriptive Analytics tells us what happened in the past or what is happening now.
It mainly answers:
"What happened?"
It takes existing data and summarizes it.

Restaurant example:
Suppose a restaurant has this monthly sales data:

Month	**Sales**
January	₹ 50,000
February	₹ 60,000
March	₹ 45,000
April	₹ 70,000

The restaurant analyzes this data.
It can say:
"Sales were ₹ 50,000 in January, ₹ 60,000 in February, ₹ 45,000 in March and ₹ 70,000 in April."
This is **Descriptive Analytics** because we are simply describing **what happened**.

Real-life examples

Descriptive Analytics can answer:

- How many products were sold?
- What was the total revenue?
- How many students passed?
- How many customers visited the website?
- What were the sales last month?

2. Diagnostic Analytics

"Descriptive Analytics told us WHAT happened. But now we want to know WHY it happened."

So Diagnostic Analytics answers:

"Why did it happen?"

Continue the same restaurant example

From our previous data:

Month	Sales
January	₹ 50,000
February	₹ 60,000
March	₹ 45,000
April	₹ 70,000

We already know that:
March had the lowest sales.
That's descriptive analytics.

Now the restaurant asks:
"Why were sales low in March?"

The company analyzes other data and discovers:

- There was heavy rain.
- Fewer customers visited the restaurant.
- One popular food item was unavailable.
- There were fewer promotional offers.

Now the company understands the **reason** for the low sales.
That is **Diagnostic Analytics**.

Real-life example

Suppose a college notices:
“Student attendance decreased in August.”
That's descriptive.

Then the college investigates and discovers:
“Attendance decreased because many students were absent during a festival period.”
That's diagnostic.

3. Predictive Analytics

“So far, we looked at the past. First , we asked WHAT happened, then WHY it happened. Now we want to look toward the future.”

Predictive Analytics answers:
“What is likely to happen in the future?”
It uses **past and current data** to make predictions about the future.

Continue our restaurant example
Suppose the restaurant has analyzed several years of sales data.

It notices:
“Whenever summer starts, cold drinks and ice creams sell much more.”
Now summer is approaching.

The restaurant may predict:
“Cold drink sales are likely to increase next month.”
This is **Predictive Analytics**.

Notice something important:
We are **not saying it will definitely happen**.
We are saying:
“Based on past data, this is likely to happen.”

Another example: Student marks

Suppose a college has data about students:

- Attendance
- Assignment marks
- Internal exam marks
- Previous exam performance

The college analyzes this data and predicts:

"Students with very low attendance and low internal marks may be at higher risk of failing the final exam."

This is predictive analytics.

"Predictive Analytics uses past and current data to predict what is likely to happen in the future."

4. Prescriptive Analytics

"We have now understood what happened, why it happened, and what may happen. But there is one more important question."

Ask:

"If we know what is likely to happen, what should we do about it?"

This is **Prescriptive Analytics**.

It answers:

"What should we do?"

Continue the restaurant example

Predictive Analytics tells us:

"Cold drink sales are likely to increase next month."

Now the restaurant has to decide:

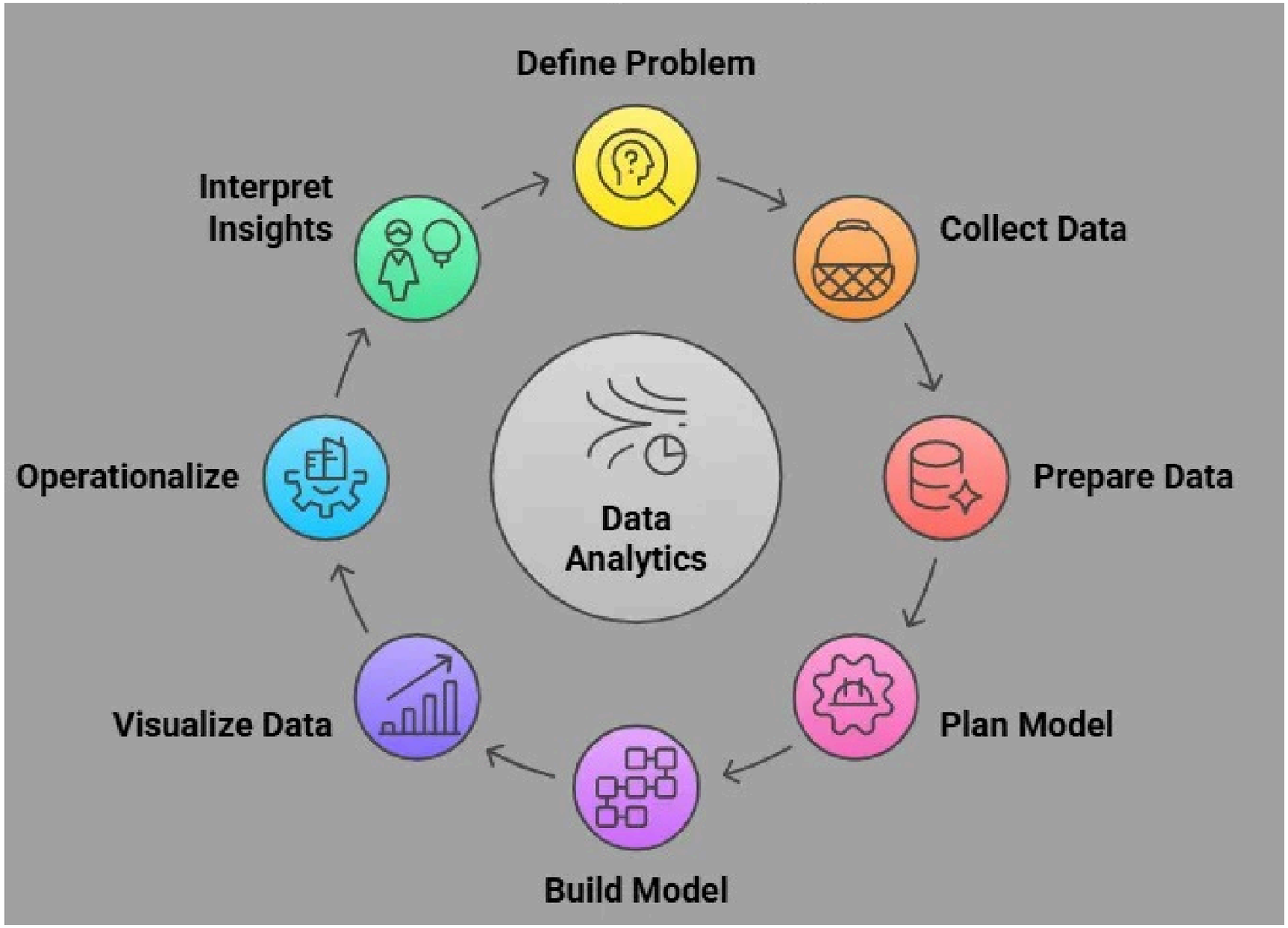
"What should we do?"

Prescriptive Analytics may suggest:

- Order more cold drinks.
- Keep extra stock.
- Give a discount on cold drinks.
- Create a combo offer.
- Increase advertising for cold drinks.

That is **Prescriptive Analytics**.

Data Analytics Lifecycle



1. Define the Problem Statement

First, we need to clearly understand **what problem we want to solve**.

We ask:

"What do we want to find out from the data?"

Example – College:

A college wants to know:

"Why are some students getting low marks?"

So, the problem statement is:

Identify the factors affecting students' academic performance.

2. Collect the Data

Once we know the problem, we need to **collect the required data**.

Data can come from:

- College databases
- Surveys
- Excel files
- Websites
- Sensors
- Applications
- Social media

Example:

For student performance, we may collect:

Student	Attendance	Study Hours	Assignment Marks	Exam Marks
A	90%	5	85	88
B	60%	2	60	55
C	80%	4	75	78

3. Prepare the Data

The collected data may not be ready for analysis.
It can contain:

- Missing values
- Duplicate records
- Incorrect values
- Different formats
- Unnecessary data

So ,we **clean and prepare the data**.

For example:
Attendance = 90%
but another record may have:
Attendance = "90 percent"
We need to convert them into a common format.

Simple meaning:
Prepare = Clean the data and make it ready for analysis

4. Plan the Model

Now we decide **how we are going to analyze the data**.

We decide:

- What variables/features are important?
- Which analytical method should we use?
- What type of model is suitable?
- What are we trying to predict or classify?

For example:
If we want to **predict whether a student will pass or fail**, we may choose a classification model such as:
Decision Tree / Logistic Regression / Random Forest

So:
Plan Model = Decide what method/model we should use.

5. Build the Model

Now we actually **apply the selected model to the data**.

For example:
We give the student data to a machine learning model.

Attendance
Study Hours
Assignment Marks
↓
ML Model
↓
Pass / Fail

The model learns patterns from the available data

6. Visualize the Data

Now we represent the results using **graphs and charts** so that they are easier to understand.

For example:

- Bar chart
- Histogram
- Pie chart
- Scatter plot
- Box plot
- Line chart

We might create a graph showing:
Attendance vs Exam Marks

This may help us visually see that students with higher attendance generally have higher marks.
So:
Visualization = Convert data/results into graphs so humans can understand them easily.

7. Interpret the Insights

This is where we ask:
"What does the result actually mean?"

Suppose our analysis shows:
Students with attendance above 80% generally score higher.
This is an **insight**.

We don't just look at the graph—we explain what the result tells us.

For example:
Insight:
"Attendance appears to have a positive relationship with academic performance."

8. Operationalize

Finally, we **put the results into practical use**.
The analysis should help someone **take action or make a decision**.

For example, the college may create an early-warning system:
Student attendance < 60%
↓
System identifies student
↓
Teacher receives notification
↓
Counselling / Support provided

So:
Operationalize = Take the analytics/model into the real world and use it for decision-making.

Types of Data

1. Structured Data

Structured data is data that is **organized in a fixed format**, usually in rows and columns.
Think about an **Excel sheet**.

Student ID	Name	Age	Marks
101	Ravi	20	85
102	Priya	21	90
103	Arun	20	78

Everything has a proper place:

- Student ID → one column
- Name → one column
- Age → one column
- Marks → one column

Examples

- Excel tables
- SQL databases
- Student records
- Bank transaction tables
- Employee records

2. Semi-Structured Data

Semi-structured data **does not follow a strict table format**, but it still contains some organization using **tags, keys, or metadata**.

Common examples are:

- JSON
- XML
- HTML

For example, JSON:

```
{
  "name": "Ravi",
  "age": 20,
  "marks": 85
}
```

There are no rows and columns like an Excel table.

But we still know:
name → Ravi
age → 20
marks → 85
So, it has **some structure**, but not a fixed tabular structure.

3. Unstructured Data

Unstructured data is data that **does not have a predefined structure**.

Examples:

- Images
- Videos
- Audio
- Emails
- Social media posts
- PDF documents
- Text documents

For example, consider a photograph:
📷 A photo does not have columns such as:
Name | Age | Marks
It is simply an image containing pixels.

Similarly:
🎵 An audio recording contains sound.
🎥 A video contains images, sound, etc.

4. Time-Series Data

Time-series data is data that is **collected or recorded over time**.
The important thing here is **time**.
For example, temperature recorded every hour:

Time	Temperature
9 AM	25° C
10 AM	27° C
11 AM	29° C
12 PM	31° C

Here, the data is connected to time.

Examples

- Stock prices
- Daily sales
- Weather temperature
- Monthly electricity usage
- Website visitors per day
- Sensor readings
- Heart-rate readings over time

Time-series data = Data collected in chronological order over time.

Data Sources

"Where does this data come from?"
That is what we mean by **data sources**.

Data sources can be broadly divided into:

A. Internal Data Sources
Data generated **inside an organization**.
Examples:
- Company database
- Sales records
- Employee records
- Customer records
- Attendance system
- Transaction records

B. External Data Sources
Data obtained **from outside the organization**.
Examples:
- Government websites
- Social media
- Public datasets
- Weather services
- Online websites
- Research organizations

Common Data Sources

Data Source	Example
Database	MySQL, PostgreSQL
Files	Excel, CSV
Websites	Web data
APIs	Weather API
Sensors	Temperature sensor
Social Media	Posts, comments
Surveys	Google Forms
Applications	Mobile app data
Cloud	Cloud databases/storage

Data Formats

Now the next question is,
"How is data stored?"
The answer is **data formats**.

Some common formats are:
CSV
CSV = Comma-Separated Values

Example:
ID,Name,Marks
101,Ravi,85
102,Priya,90
103,Arun,78
Usually used for **tabular/structured data**.

Excel
Example:
Student.xlsx
Contains rows, columns, formulas, multiple sheets, etc.
Used commonly for structured data.

JSON
Example:

```
{
  "name": "Ravi",
  "age": 20,
  "marks": 85
}
```

Commonly used for **web applications and APIs**.

XML
Example:
<student>
  <name>Ravi</name>
  <age>20</age>
  <marks>85</marks>
</student>
Uses **tags** to organize information.

Image
Examples:
.jpg
.png
.jpeg
Contains visual information.

Text
Examples:
.txt
.docx
.pdf
Can contain written information such as reports, documents, and messages.

Audio
Examples:
.mp3
.wav
Contains sound.

Video
Examples:
.mp4
.avi
.mkv
Contains video information, often along with audio.

Python for data analytics: NumPy, Pandas, reading CSV/JSON files, indexing, filtering, grouping and aggregation.

1. NumPy
2. Pandas
3. Reading CSV/JSON files
4. Indexing
5. Filtering
6. Grouping
7. Aggregation

"Suppose you are working as a Data Analyst in an e-commerce company. The company has a large amount of sales data. The management wants to know which products are selling well, which region has the highest sales, whether there are missing or duplicate records, and how sales information can be combined with product and regional information. “

"Today we will learn how Python can help us answer these business questions using NumPy and Pandas."

NUMPY

It stands for Numerical Python, NumPy helps us perform calculations on numerical data efficiently. (OR)

NumPy is a Python library used for efficient numerical computing and working with arrays.

NumPy
↓
Numerical Python
↓
Work with numerical data

Uses

NumPy is used for:

- Creating numerical arrays
- Mathematical calculations
- Array operations
- Statistical calculations

For example:
Sales = [50000, 5000, 8000, 12000]

Questions:

- What is the total sales?
- What is the average sales?
- What is the highest sales?

```python
import numpy as np


# Create a NumPy Array

sales = np.array([50000, 2500, 9000, 30000, 55000])

print("NUMPY ARRAY:")
print(sales)


# Indexing

print("\nFIRST SALES VALUE:")
print(sales[0])

print("\nLAST SALES VALUE:")
print(sales[-1])


# Slicing

print("\nFIRST THREE SALES VALUES:")
print(sales[0:3])


# Array Operations

print("\nSALES + 1000:")
print(sales + 1000)

print("\nSALES * 2:")
print(sales * 2)
```

```python
# Mathematical Operations

print("\nTOTAL SALES:")
print(np.sum(sales))

print("\nAVERAGE SALES:")
print(np.mean(sales))

print("\nMAXIMUM SALES:")
print(np.max(sales))

print("\nMINIMUM SALES:")
print(np.min(sales))
```

OUTPUT

```
NUMPY ARRAY:
[50000  2500  9000 30000 55000]

FIRST SALES VALUE:
50000

LAST SALES VALUE:
55000

FIRST THREE SALES VALUES:
[50000  2500  9000]

SALES + 1000:
[51000  3500 10000 31000 56000]

SALES * 2:
[100000   5000  18000  60000 110000]

TOTAL SALES:
146500

AVERAGE SALES:
29300.0

MAXIMUM SALES:
55000

MINIMUM SALES:
2500
```

"NumPy allows us to perform calculations on the entire array without writing a loop for every element."

Then transition:
"But our real sales data contains not only numbers. It contains product names, categories, regions, dates, sales and profit. For this type of tabular data, Pandas is more useful."

Pandas

Pandas is a Python library used to store, manipulate, analyze and visualize structured or tabular data.

Introduce two important structures:

Series

A single column of data. (or) "A Series is similar to one column in a table."

DataFrame

A complete table containing rows and columns.

Coding – Series

```
import pandas as pd


marks = pd.Series([80, 75, 90, 85])

print("PANDAS SERIES:")
print(marks)
```

```
PANDAS SERIES:
0    80
1    75
2    90
3    85
dtype: int64
```

Coding – DataFrame

```
data = {
    "Name": ["Asha", "Ravi", "Meera"],
    "Marks": [88, 72, 91],
    "Branch": ["CSE", "ECE", "CSE"]
}

df = pd.DataFrame(data)

print("PANDAS DATAFRAME:")
print(df)
```

```
PANDAS DATAFRAME:
    Name  Marks Branch
0   Asha     88    CSE
1   Ravi     72    ECE
2  Meera     91    CSE
```

"A DataFrame is a two-dimensional table consisting of rows and columns."

"Instead of creating our sales data manually, suppose the company gives us a CSV file."

Reading a Real CSV Dataset

CSV stands for Comma-Separated Values. It is one of the most commonly used formats for storing tabular data.

Pandas provides:
pd.read_csv() to read CSV files.

```
import pandas as pd


df = pd.read_csv("sales_data.csv")


print("ORIGINAL SALES DATASET:")
print(df)
```

```
ORIGINAL SALES DATASET:
      Order Date Product Name     Category Region  Quantity  Sales   Profit
0     2024-12-31      Printer       Office  North         4   3640   348.93
1     2022-11-27        Mouse  Accessories   East         7   1197   106.53
2     2022-05-11       Tablet  Electronics  South         5   5865   502.73
3     2024-03-16        Mouse  Accessories  South         2    786   202.87
4     2022-09-10        Mouse  Accessories   West         1    509   103.28
...          ...          ...          ...    ...       ...    ...      ...
3495  2023-02-15      Monitor  Accessories  North         4   4064   771.16
3496  2022-09-18      Monitor  Accessories   East         1   1117   119.89
3497  2022-04-12       Laptop  Electronics  South         4    260    66.02
3498  2022-01-18      Printer       Office  South         3    222    50.28
3499  2024-11-01   Smartwatch  Electronics   East         9  10530  1777.32

[3500 rows x 7 columns]
```

Reading JSON Files

JSON is another commonly used format for storing and exchanging data, especially between applications and APIs.

Coding

```
json_df = pd.read_json("sales_data.json")

print("JSON DATASET:")
print(json_df)
```

CSV → pd.read_csv()
JSON → pd.read_json()

Understanding the Dataset

"After loading the data, a data analyst should never immediately start making conclusions. First, we need to understand the dataset."

1. First 5 Records

head() displays the first five rows by default.

"This gives us a quick look at what the dataset contains."

```
print("FIRST 5 RECORDS:")
print(df.head())
```

```
FIRST 5 RECORDS:
   Order Date Product Name     Category Region  Quantity  Sales  Profit
0  2024-12-31      Printer       Office  North         4   3640  348.93
1  2022-11-27        Mouse  Accessories   East         7   1197  106.53
2  2022-05-11       Tablet  Electronics  South         5   5865  502.73
3  2024-03-16        Mouse  Accessories  South         2    786  202.87
4  2022-09-10        Mouse  Accessories   West         1    509  103.28
```

2. Last 5 Records

"tail() displays the last five records."

```
print("LAST 5 RECORDS:")
print(df.tail())
```

```
LAST 5 RECORDS:
      Order Date Product Name     Category Region  Quantity  Sales   Profit
3495  2023-02-15      Monitor  Accessories  North         4   4064   771.16
3496  2022-09-18      Monitor  Accessories   East         1   1117   119.89
3497  2022-04-12       Laptop  Electronics  South         4    260    66.02
3498  2022-01-18      Printer       Office  South         3    222    50.28
3499  2024-11-01   Smartwatch  Electronics   East         9  10530  1777.32
```

3. Dataset Shape

```
print("DATASET SHAPE:")
print(df.shape)
```

```
DATASET SHAPE:
(3500, 7)
```

4. Column Names

```
print("COLUMN NAMES:")
print(df.columns)
```

```
COLUMN NAMES:
Index(['Order Date', 'Product Name', 'Category', 'Region', 'Quantity', 'Sales',
       'Profit'],
      dtype='str')
```

"This tells us what information is available in our dataset."

5. Data Types

"This tells us whether each column contains integers, decimal values, strings, dates, etc."

```
print("DATA TYPES:")
print(df.dtypes)
```

```
DATA TYPES:
Order Date          str
Product Name        str
Category            str
Region              str
Quantity          int64
Sales             int64
Profit          float64
dtype: object
```

Indexing

Indexing means selecting specific rows, columns or values from a DataFrame.

"Suppose the manager doesn't want the entire dataset. She only wants the Product Name column."

Column Indexing

```
print("PRODUCT NAME:")
print(df["Product Name"])
```

```
PRODUCT NAME:
0          Printer
1            Mouse
2           Tablet
3            Mouse
4            Mouse
           ...
3495       Monitor
3496       Monitor
3497        Laptop
3498       Printer
3499    Smartwatch
Name: Product Name, Length: 3500, dtype: str
```

Multiple Columns

We can select more than one column by giving the column names inside a list."

```
print("PRODUCT AND SALES:")
print(df[["Product Name", "Sales"]])
```

```
PRODUCT AND SALES:
     Product Name  Sales
0         Printer   3640
1           Mouse   1197
2          Tablet   5865
3           Mouse    786
4           Mouse    509
...           ...    ...
3495      Monitor   4064
3496      Monitor   1117
3497       Laptop    260
3498      Printer    222
3499   Smartwatch  10530

[3500 rows x 2 columns]
```

Row Indexing

```
print("FIRST ROW:")
print(df.iloc[0])
```

```
FIRST ROW:
Order Date      2024-12-31
Product Name       Printer
Category            Office
Region               North
Quantity                 4
Sales                 3640
Profit              348.93
```

"iloc selects data based on its numerical position."

Specific Value

```
print("SALES OF FIRST RECORD:")
print(df.iloc[0, 5])
```

```
SALES OF FIRST RECORD:
3640
```

"This gives the value located at that row and column position."

Slicing (when you want to extract a group of elements)

```
print("FIRST THREE ROWS:")
print(df.iloc[0:3])
```

```
FIRST THREE ROWS:
   Order Date Product Name     Category Region  Quantity  Sales  Profit
0  2024-12-31      Printer       Office  North         4   3640  348.93
1  2022-11-27        Mouse  Accessories   East         7   1197  106.53
2  2022-05-11       Tablet  Electronics  South         5   5865  502.73
```

```
print("FIRST THREE ROWS:")
print(df.iloc[3:])
```

```
FIRST THREE ROWS:
      Order Date Product Name     Category Region  Quantity  Sales   Profit
3     2024-03-16        Mouse  Accessories  South         2    786   202.87
4     2022-09-10        Mouse  Accessories   West         1    509   103.28
5     2023-12-01       Camera  Electronics   West         1    524   106.35
6     2023-10-09   Headphones  Accessories  North         7   6167  1027.98
7     2022-01-14       Camera  Electronics  South         7   3059   873.50
...          ...          ...          ...    ...       ...    ...      ...
3495  2023-02-15      Monitor  Accessories  North         4   4064   771.16
3496  2022-09-18      Monitor  Accessories   East         1   1117   119.89
3497  2022-04-12       Laptop  Electronics  South         4    260    66.02
3498  2022-01-18      Printer       Office  South         3    222    50.28
3499  2024-11-01   Smartwatch  Electronics   East         9  10530  1777.32

[3497 rows x 7 columns]
```

```
print("FIRST THREE ROWS:")
print(df.iloc[:2])
```

```
FIRST THREE ROWS:
   Order Date Product Name     Category Region  Quantity  Sales  Profit
0  2024-12-31      Printer       Office  North         4   3640  348.93
1  2022-11-27        Mouse  Accessories   East         7   1197  106.53
```

Filtering

"The manager wants to see only the transactions where sales are greater than ₹ 10,000."

Filtering means selecting only the rows that satisfy a particular condition.

```
df["Sales"] > 10000
```

checks every row.

```
True  → keep the row
False → remove the row
```

```
result = df[df["Sales"] > 10000]

print("SALES GREATER THAN 10000:")
print(result)
```

```
SALES GREATER THAN 10000:
      Order Date Product Name     Category Region  Quantity  Sales   Profit
32    2023-04-09       Laptop  Electronics  North         9  10773  1907.36
147   2022-10-23     Keyboard  Accessories   West         9  10035  1486.30
153   2024-08-08      Monitor  Accessories  North         9  10530  2387.63
267   2024-01-25       Tablet  Electronics   East         9  10188  2165.78
293   2023-08-08      Monitor  Accessories  North         9  10458  2799.48
379   2022-05-09      Printer       Office   West         9  10656   961.30
846   2024-05-14       Camera  Electronics   West         9  10143  1690.66
1126  2024-04-21   Smartphone  Electronics  South         9  10314  2729.60
1127  2023-09-29     Keyboard  Accessories  South         9  10782  2730.14
1430  2024-03-04   Smartphone  Electronics   West         9  10458  1835.21
1489  2023-12-18      Monitor  Accessories   West         9  10665  1595.08
1540  2023-05-18   Headphones  Accessories   West         9  10134  1334.37
1620  2023-01-28   Smartwatch  Electronics   West         9  10206   592.77
1967  2023-12-07   Smartphone  Electronics   West         9  10404  2028.21
2129  2024-01-23       Camera  Electronics  North         9  10188  1408.18
2144  2023-12-06        Mouse  Accessories  North         9  10593  2488.69
2347  2022-11-16     Keyboard  Accessories  South         9  10629  2843.54
2385  2023-04-02      Monitor  Accessories   West         9  10413  1738.63
2392  2023-02-02       Laptop  Electronics  South         9  10620  2946.93
2573  2022-06-27       Laptop  Electronics   East         9  10728  1967.84
3019  2022-09-12   Smartwatch  Electronics  North         9  10251   737.45
3179  2023-08-29   Headphones  Accessories  South         9  10080  1794.73
3476  2022-07-29       Camera  Electronics   West         9  10017  1468.72
3499  2024-11-01   Smartwatch  Electronics   East         9  10530  1777.32
```

Kavitha S

Multiple Conditions

"Show North region transactions where sales are greater than ₹ 10,000."

```
result = df[
    (df["Region"] == "North") &
    (df["Sales"] > 10000)
]

print("NORTH REGION SALES GREATER THAN 10000:")
print(result)
```

```
NORTH REGION SALES GREATER THAN 10000:
      Order Date Product Name     Category Region  Quantity  Sales   Profit
32    2023-04-09       Laptop  Electronics  North         9  10773  1907.36
153   2024-08-08      Monitor  Accessories  North         9  10530  2387.63
293   2023-08-08      Monitor  Accessories  North         9  10458  2799.48
2129  2024-01-23       Camera  Electronics  North         9  10188  1408.18
2144  2023-12-06        Mouse  Accessories  North         9  10593  2488.69
3019  2022-09-12   Smartwatch  Electronics  North         9  10251   737.45
```

Exploratory Data Analysis

"Exploratory Data Analysis, or EDA, means examining and understanding the dataset before performing further analysis. "

EDA helps us:

- Understand the data
- Find patterns
- Find missing values
- Find duplicate records
- Understand numerical values

Descriptive Statistics

describe() gives statistical information such as:

- Count
- Mean
- Standard deviation
- Minimum
- 25%
- 50%
- 75%
- Maximum

```
print("DESCRIPTIVE STATISTICS:")
print(df.describe())
```

```
DESCRIPTIVE STATISTICS:
          Quantity         Sales       Profit
count  3500.000000   3500.000000  3500.000000
mean      4.931714   3047.966000   527.047203
std       2.575895   2440.213237   504.139732
min       1.000000     51.000000     6.970000
25%       3.000000   1049.500000   158.695000
50%       5.000000   2350.500000   361.070000
75%       7.000000   4537.000000   729.125000
max       9.000000  10782.000000  2946.930000
```

"This helps us understand the distribution of numerical columns such as Quantity, Sales and Profit."

Missing Values

"isnull() checks whether a value is missing."
"sum() counts how many missing values are present in each column."

```
print("MISSING VALUES:")
print(df.isnull().sum())
```

```
MISSING VALUES:
Order Date      0
Product Name    0
Category        0
Region          0
Quantity        0
Sales           0
Profit          0
dtype: int64
```

Display Rows With Missing Values

"This shows the complete records where at least one value is missing."

```
print("ROWS WITH MISSING VALUES:")
print(df[df.isnull().any(axis=1)])
```

```
ROWS WITH MISSING VALUES:
Empty DataFrame
Columns: [Order Date, Product Name, Category, Region, Quantity, Sales, Profit]
Index: []
```

Duplicate Records

```
print("TOTAL DUPLICATE RECORDS:")
print(df.duplicated().sum())
```

"duplicated() identifies repeated rows."

```
TOTAL DUPLICATE RECORDS:
0
```

```
print("DUPLICATE RECORDS:")
print(df[df.duplicated()])
```

This displays the actual duplicate records.

```
DUPLICATE RECORDS:
Empty DataFrame
Columns: [Order Date, Product Name, Category, Region, Quantity, Sales, Profit]
Index: []
```

Grouping and Aggregation

"Suppose the manager doesn't want to see every transaction. She wants to know the total sales for each region. "

Grouping means dividing the data into groups based on a particular column.
Aggregation means performing a calculation such as sum, mean, maximum or minimum on each group.

Total Sales by Region

```
region_sales = df.groupby("Region")["Sales"].sum()

print("TOTAL SALES BY REGION:")
print(region_sales)
```

```
TOTAL SALES BY REGION:
Region
East     2675110
North    2488773
South    2659548
West     2844450
```

```
groupby("Region")
        ↓
Create groups for North, South, East, West
        ↓
["Sales"]
        ↓
Select Sales column
        ↓
sum()
        ↓
Calculate total sales for each region
```

Average Sales by Region

```
region_average = df.groupby("Region")["Sales"].mean()

print("AVERAGE SALES BY REGION:")
print(region_average)
```

```
AVERAGE SALES BY REGION:
Region
East     3106.980256
North    2900.667832
South    3011.945640
West     3167.538976
Name: Sales, dtype: float64
```

Total Profit by Region

```
region_profit = df.groupby("Region")["Profit"].sum()

print("TOTAL PROFIT BY REGION:")
print(region_profit)
```

```
TOTAL PROFIT BY REGION:
Region
East     464888.46
North    426314.75
South    458103.27
West     495358.73
```

Multiple Aggregations Together

```
region_summary = df.groupby("Region")["Sales"].agg(
    ["sum", "mean", "max", "min", "count"]
)

print("SALES SUMMARY BY REGION:")
print(region_summary)
```

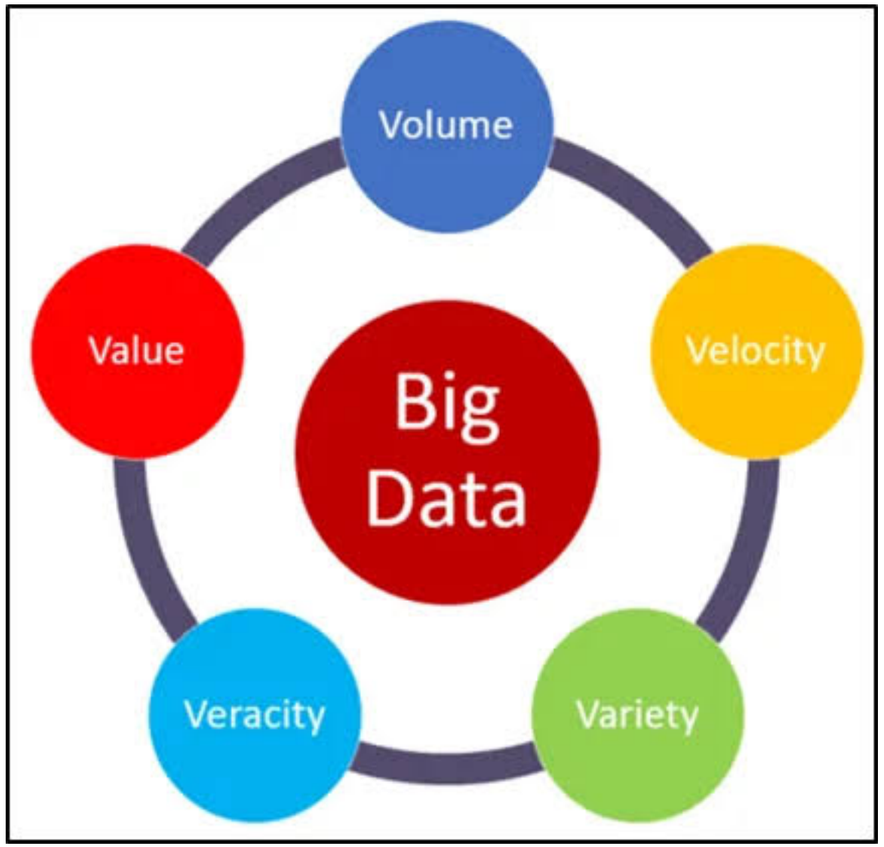
```
SALES SUMMARY BY REGION:
             sum         mean    max  min  count
Region
East     2675110  3106.980256  10728   58    861
North    2488773  2900.667832  10773   58    858
South    2659548  3011.945640  10782   54    883
West     2844450  3167.538976  10665   51    898
```

Overview of Big Data: characteristics (5Vs), ecosystem and tools

What is Big Data?

- Big Data refers to extremely large and complex amounts of data that are difficult to store, process, and analyze using traditional data-processing systems. (OR)
- "Big Data is not simply data that is big. It is data that is very large, generated very quickly, comes in different forms, may have quality problems, and can provide useful information when properly analyzed."

The 5Vs of Big Data (OR) Characteristics of Big Data — 5Vs



The five characteristics are:

1.Volume
2.Velocity
3.Variety
4.Veracity
5.Value

Volume

Volume refers to the enormous quantity of data generated, collected, and stored by organizations and digital systems.

Traditional databases are generally designed to handle relatively smaller amounts of data. Big Data environments are designed to handle data ranging from **gigabytes (GB) to terabytes (TB), petabytes (PB), and beyond**.

Sources of Large Data Volumes

Data can be generated from:

- Social media platforms
- E-commerce transactions
- Banking systems
- Websites and mobile applications
- IoT devices
- Sensors
- Video and audio platforms
- Business transactions
- Machine and application logs

Example — E-Commerce

Consider an online shopping platform.

Every customer interaction generates data:

Customer Registration

↓

Product Search

↓

Product View

↓

Add to Cart

↓

Purchase

↓

Payment

↓

Product Review

When millions of customers perform these activities, the organization generates an enormous volume of data.

Volume answers the question: "How much data is being generated or stored?"

Velocity

Velocity refers to the speed at which data is generated, transmitted, collected, and processed.

In many modern applications, data is generated continuously and needs to be processed quickly or in real time.

Examples of High-Velocity Data

- Online banking transactions
- Stock market transactions
- Social media activities
- IoT sensor readings
- GPS location updates
- Website clicks
- Online advertisements
- Real-time monitoring systems

Example — Online Banking
Suppose thousands of customers perform transactions at the same time.

Customer A → ₹ 5,000 transaction
Customer B → ₹ 2,000 transaction
Customer C → ₹ 10,000 transaction
Customer D → ₹ 7,000 transaction
↓
Data generated continuously
↓
Banking system processes the transactions

The system must process this data quickly to:

- Confirm transactions
- Detect suspicious activity
- Update account balances
- Generate notifications

Variety

Variety refers to the different types, formats, and structures of data generated from various sources.

Unlike traditional systems that mainly work with structured tables, Big Data can contain **structured, semi-structured, and unstructured data**.

Types of Data

1. Structured Data

Data organized in a predefined format, usually rows and columns.

Example:

Customer_ID	Name	Age	Purchase
101	Ravi	25	Laptop
102	Anu	28	Mobile

Examples:

- Relational databases
- Spreadsheets
- Transaction records

2. Semi-Structured Data
Data that does not follow a strict table structure but contains tags, keys, or other organizational elements.

Examples:
- JSON
- XML
- HTML
- Application logs

Example:
```
{
   "customer_id": 101,
   "product": "Laptop",
   "price": 50000
}
```

3. Unstructured Data
Data that does not have a predefined data structure.

Examples:
- Images
- Videos
- Audio
- Emails
- PDF documents
- Social media posts
- Text documents

Example — Social Media

A social media platform can receive:

Text
↓
"Great product!"

Image
↓
Product photograph

Video
↓
Product demonstration

Audio
↓
Voice message

Location
↓
GPS coordinates

All these represent different forms of data.

Variety answers the question: "What different types and formats of data are being generated?"

Veracity

Veracity refers to the quality, accuracy, consistency, reliability, and trustworthiness of data.

Veracity answers the question: "How accurate, reliable, and trustworthy is the data?"

Big Data can contain large amounts of inaccurate, incomplete, duplicated, or inconsistent information. Therefore, organizations need to evaluate and clean data before using it for analysis and decision-making.

Common Data Quality Problems

Big Data may contain:

- Missing values
- Incorrect values
- Duplicate records
- Inconsistent information
- Outdated information
- Fake information
- Noisy data

Example — Customer Data

Suppose a database contains:

Customer	Age
Ravi	25
Anu	30
Kumar	250
Priya	—

Here:

- **250** may be an incorrect age/data.
- **—** represents missing data.

If the organization uses this data without checking it, the analysis may produce incorrect results.

Example — Product Reviews

Suppose an online shopping website has:

100 genuine reviews

+

50 fake reviews

↓

Review Analysis

If fake reviews are not identified, the company may make incorrect decisions about product quality.

Value

Value refers to the useful information, insights, and benefits that organizations obtain by analyzing Big Data.

Simply collecting and storing a huge amount of data does not provide benefits by itself.

The real objective is to **extract meaningful insights from data and use them for decision-making**.

Example — E-Commerce

Suppose an e-commerce company analyzes millions of customer transactions.

It discovers:

Customer purchase history

↓

Analysis

↓

Customers frequently buy laptops + laptop bags

↓

Product recommendation

↓

Increased sales

The useful business outcome represents **Value**.

Example — Banking
A bank analyzes transaction data and identifies unusual patterns.
Transaction Data
↓
Big Data Analysis
↓
Unusual transaction detected
↓
Possible fraud
↓
Alert generated
The ability to detect fraud provides **value** to the organization.

Value answers the question: "What useful information or benefit can be obtained from the data?"

Big Data Ecosystem and Tools

The Big Data ecosystem is not a single tool. It is a collection of technologies that work together to handle Big Data from its source to the final decision.

A **Big Data Ecosystem** is a collection of technologies, tools, platforms, and frameworks used to **collect, store, process, analyze, and visualize large and complex datasets**.

In simple words:

- **Big Data Ecosystem = Different tools working together to manage Big Data.**

Why do we need an ecosystem?

Big Data has the 5Vs:
- Large volume
- High velocity
- Variety of data
- Data quality challenges
- Need to extract value

Therefore, a single tool is generally not enough.
Different tools are used for different stages.

BIG DATA ECOSYSTEM

1. DATA SOURCES

- Websites / Web Applications
- Mobile Apps
- Social Media
- Sensors / IoT
- Transactions (E-commerce, Banking, etc.)
- Logs / Click Stream
- Other Business Applications

2. DATA INGESTION

Collect and ingest data from multiple sources (Batch / Real-time)

- Apache Kafka
- Apache NiFi
- Apache Flume

3. DATA STORAGE

Store large volume of structured, semi-structured and unstructured data

- **HDFS** (Hadoop Distributed File System)
- **Cloud Storage** (Amazon S3, Azure Blob, etc.)
- **NoSQL Databases** (HBase, Cassandra, MongoDB, etc.)

4. DATA PROCESSING

Process and transform data for analysis

- Hadoop MapReduce
- Apache Spark
- Apache Flink

5. DATA ANALYSIS

Analyze data to extract patterns, trends and useful insights

- Hive (SQL-like Query)
- Spark SQL
- Python
- R
- Machine Learning

6. DATA VISUALIZATION

Visualize and present insights through dashboards and reports

- Power BI
- Tableau
- Excel
- Custom Dashboards

7. DECISION MAKING

Use insights to make better and informed business decisions

- Improve Customer Experience
- Detect Fraud
- Increase Revenue
- Optimize Operations
- Manage Inventory
- Strategic Decision Making

ENABLING TECHNOLOGIES (Across the Ecosystem)

- Security
- Governance
- Data Quality
- Monitoring
- Metadata Management

FLOW OF DATA

Data Generated → Data Collected → Data Ingested → Data Stored → Data Processed → Data Analyzed → Data Visualized → Decision Made

Layer 1 — Data Sources

Data Sources are the places or systems from which data is generated.

In simple words:

Data Sources = Where does the data come from?

Before doing anything with Big Data, we need data.

Examples of Data Sources

Data can come from:

- Websites
- Mobile applications
- Social media
- Sensors
- IoT devices
- Banking systems
- E-commerce transactions
- GPS devices
- Application logs

E-commerce Example

Suppose a customer opens an online shopping app.

The customer:

Searches for a laptop

↓

Views a product

↓

Adds it to cart

↓

Makes payment

↓

Writes a review

Every activity generates data.

So, the **website and mobile application are data sources**.

Layer 2 — Data Ingestion

"We have data coming from websites, mobile apps, sensors, and other sources. How do we bring all this data into our Big Data system?"

Answer:

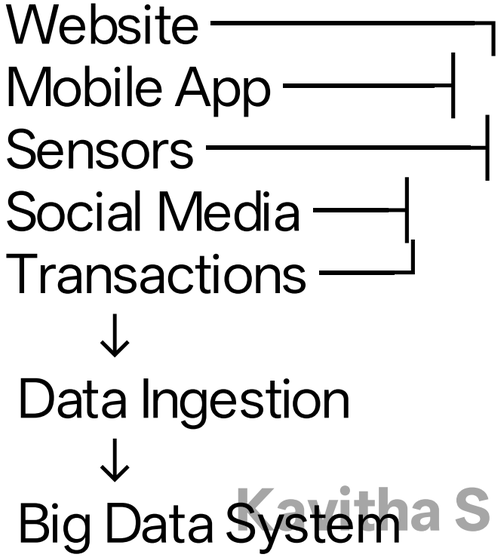
Data Ingestion.

Data ingestion is the process of collecting and transferring data from different sources into a system for storage and processing.

In simple words:

Data Ingestion = Bringing data into the Big Data system.

Example



Tools

Common tools include:

- **Apache Kafka**
- **Apache NiFi**
- **Apache Flume**

Layer 3 — Data Storage

“After collecting all this data, where do we keep it?”

Answer:

Data Storage.

Data storage is the process of storing large amounts of data so that it can be accessed and processed when required.

In simple words:

Data Storage = Where do we keep the data?

Examples

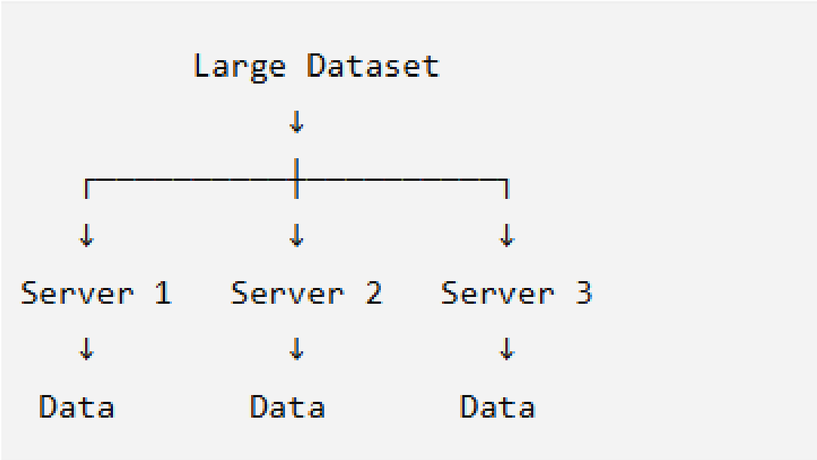
- HDFS
- Cloud storage
- Amazon S3
- NoSQL databases
- Data lakes

HDFS Example

HDFS stands for:

Hadoop Distributed File System

It stores large datasets across multiple machines.



This is called **distributed storage**.

Layer 4 — Data Processing

“We have stored billions of records. Can we directly analyze all this raw data?”

Usually, we first need to **process and transform** it.

Data processing is the process of transforming raw data into a useful and organized form for analysis.

In simple words:

Data Processing = Convert raw data into usable data.

Example

Suppose we have raw transactions:

Order 101 → Laptop → ₹ 50,000
Order 102 → Phone → ₹ 30,000
Order 103 → Laptop → ₹ 50,000
Order 104 → Tablet → ₹ 20,000

Processing can help us calculate:

Laptop → 2 orders
Phone → 1 order
Tablet → 1 order

We can also:

- Remove duplicate records
- Handle missing values
- Convert data formats
- Filter unwanted data
- Aggregate data
- Calculate totals and averages

Tools

Major tools include:

- **Hadoop MapReduce**
- **Apache Spark**
- **Apache Flink**

Layer 5 — Data Analysis

After processing the data,

"What can we learn from this data?"

This is **Data Analysis**.

Data analysis is the process of examining processed data to identify patterns, trends, relationships, and useful information.

In simple words:

Data Analysis = Find useful information from the data.

E-commerce Example

The company may ask:

Which product sells the most?

Which region has the highest sales?

Which customers purchase frequently?

Which products are often purchased together?

Which month has the highest sales?

Suppose the analysis shows:

Laptop

↓

Highest-selling product

The company now has useful information.

Tools

- SQL
- Hive
- Spark SQL
- Python
- R
- Machine Learning

Layer 6 — Data Visualization

"We have found useful information. How can we show it to managers or users in an easy-to-understand way?"

Answer:

Data Visualization.

Data visualization is the graphical representation of data using charts, graphs, dashboards, and other visual formats.

In simple words:

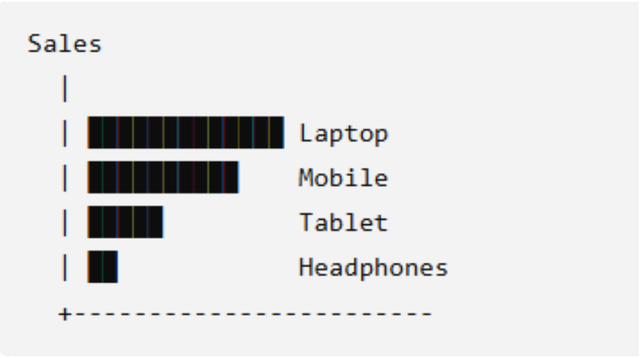
Data Visualization = Show data in a visual and understandable form.

Example

Suppose analysis gives:

Laptop ₹ 50,00,000
Mobile ₹ 40,00,000
Tablet ₹ 20,00,000
Headphones ₹ 10,00,000

Instead of giving the manager a large table, we can create a bar chart:



Now it is easy to identify the best-selling category.

Tools

- Power BI
- Tableau
- Python
- Excel

Layer 7 — Decision Making

This is the **final objective**.

"Why did we collect, store, process, and analyze all this data?"

Answer:

To make better decisions.

Decision making is the process of using insights obtained from data to make informed business or operational decisions.

In simple words:

Decision Making = Use the information to decide what to do next.

E-commerce Example

Analysis shows:

Laptop sales are increasing

↓

Company gets the insight

↓

Decision:

Increase laptop inventory

↓

Run laptop promotions

If data shows that laptop demand is increasing, the company can increase inventory and promotional activities.

END OF MODULE -01